

AVISHEK CHOUDHURY

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EDUCATION

University Of Utah , Salt Lake City, UT, US MASTER OF SCIENCE IN COMPUTING Graduate Algorithms, Distributed Systems, Artificial Intelligence, Machine Learning, Natural Language Processing	Aug 2023 - May 2025 GPA: 3.87/4
University Of Utah , Salt Lake City, UT, US MASTERS, SOFTWARE DEVELOPMENT Operating Systems, Networks and Security, Software Engineering (transferred to Computing with Certificate)	Aug 2022 - May 2023 GPA: 4.00/4

TECHNICAL SKILLS

Programming Languages: Java, C++, Go, C#, Python, JavaScript, SQL
Technologies: .NET, Spring, RESTful APIs, GraphQL, NoSQL, gRPC, Redis, RabbitMQ, Kafka, OAuth, JWT, Unit Testing, Postman, React
Cloud & Infrastructure: Azure, Kubernetes, Docker, Mircoservices, CI/CD, Linux/Unix, Git, Observability, Windows, macOS, Shell, TCP/IP
ML, AI & Data: MSSQL, PostgreSQL, MySQL, Qdrant, MongoDB, Spark, RAG, Vector Search, LLM, PyTorch, TensorFlow, Hugging Face

WORK EXPERIENCE

University Of Utah Health	Salt Lake City, UT, US
Software Engineer C#, .NET, REST APIs, Redis, RabbitMQ, Azure DevOps, SQL, NoSQL, Qdrant, OpenAI API, JWT, React - Centralized authentication and session management across 8+ distributed services by integrating OpenID Connect (OIDC) on the API Gateway. - Automated entity and relationship discovery across 1K+ SQL tables by architecting a RAG pipeline using vector search and context retrieval. - Scaled and secured a RAG platform with caching, rate limiting, and asynchronous queuing to absorb traffic bursts of 2K+ requests/second. - Reduced clinical research effort by 20% by building an LLM based SQL platform, giving conversational access to large-scale healthcare datasets.	Dec 2025 – Present
University Of Utah	Salt Lake City, UT, US
Teaching Assistant and Researcher Operating Systems, Database Systems, C++, C#, Python, PyTorch, ML - Mentored 50+ M.S. students in Operating and Database Systems, providing technical guidance, reviewing code, and resolving complex issues. Open sourced a medical Q&A LLM, reaching 5K+ downloads at Hugging Face by fine-tuning Meta's Llama model using LoRA PEFT approach.	Aug 2023 – May 2025
Software Development Intern Python, PyTorch, Spark, Google Analytics, PostgreSQL - Expanded consumer base by 10% by designing a product recommendation system for an e-commerce platform, partnering with the sales team. - Optimized ad placement strategy by analyzing user traffic data, leading to a 20% increase in click-through rates and campaign effectiveness.	Jan 2023 – Aug 2023
Accenture	Hyderabad, India
Senior Software Engineer C#, .NET Core, React, RESTful APIs, Redis, RabbitMQ, JWT, Unit Test, SQL, Microsoft Azure - Delivered and owned a UK project expected to generate \$1.2M revenue by transforming business requirements into scalable software solutions. - Scaled data exchange across distributed services by designing a fault-tolerant asynchronous messaging architecture processing 5K+ events/sec. - Improved average response time by 10% by implementing cache and rate limiting on the API layer and code optimization in the business layer. - Reduced policy creation time by 5 minutes by designing a buyer-policy matching algorithm to automate buyer search over 500K records.	Nov 2018 – Jul 2022

PROJECTS

Distributed Key-Value Store Using Raft Consensus Algorithm Golang, Unix, Multi-threading, gRPC Engineered a fault-tolerant key-value store with strong consistency and persistent recovery, achieving automatic leader failover within 4 seconds.	Code
Real-time Group Chat Server Java, WebSocket, Muti-threading, TCP/IP, HTTPS Engineered a concurrent chat server sustaining 15K+ requests/sec in load testing, using connection pooling and multi-threaded processing.	Code
Caching DNS Resolver Java, Wireshark, Muti-threading, Caching, Unix, TCP/IP, HTTPS Benchmarked 20K+ RPS using caching and request pooling on cache miss by fetching the IP details of a new domain from Google DNS server.	Code

LEADERSHIP & ACHIEVEMENTS

- Received 100% **tuition benefit** for the MS in Computing program at the University of Utah for the **Teaching Assistant** role.
- Received recognition** (given to 15% employees) twice for delivering two high value projects and knowledge transfer sessions at Accenture.
- Certifications:** [Deep Learning Specialization](#), [Microsoft Azure Fundamentals](#), [Azure Developer Associate](#), [ReactJS](#), [JavaScript](#)